

# Edoardo Vittori, PhD

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## PROFESSIONAL EXPERIENCE

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 2022 - PRESENT | <b>Intesa Sanpaolo IMICIB, Milan - Vice President, AI Investments</b> <ul style="list-style-type: none"><li>• Conducting quantitative research for futures and FX with machine learning</li><li>• Researching alpha signals from limit order book and trades and quotes data</li><li>• Supervising strategy implementation into production and monitoring live trading</li><li>• Researching daily systematic cross asset investment strategies with Epsilon SGR</li><li>• Active in the ML in finance research community with publications and paper reviews for conferences</li></ul> |
| 2018 - 2022    | <b>Intesa Sanpaolo IMICIB, Milan - Associate, XVA Management</b> <ul style="list-style-type: none"><li>• Pricing XVAs of derivatives and hedging XVA book</li><li>• Conducting quantitative research and academic research as a doctoral student</li><li>• Job rotation 2020 - Epsilon SGR Asset Management: quantitative investing</li><li>• Job rotation 2019 - Intesa Sanpaolo: credit flow trading</li></ul>                                                                                                                                                                        |
| 2016 - 2018    | <b>Banca IMI, Milan - Analyst, XVA Management</b> <ul style="list-style-type: none"><li>• Pricing the XVAs of derivatives and hedging the XVA book</li><li>• Trading of IRS, FX spot, equity futures and options, fixed income futures, CDS indexes</li><li>• Working closely with sales, trading, structuring and risk teams to structured derivatives for clients</li></ul>                                                                                                                                                                                                           |
| SUMMER 2015    | <b>AMS (Alpha Magnetic Spectrometer), CERN, Geneva - Intern</b> <ul style="list-style-type: none"><li>• Analyzing data from AMS under the supervision of Nobel prize Samuel C. Ting</li><li>• Coordinating with NASA to monitor experiments and subsystems on the International Space Station</li></ul>                                                                                                                                                                                                                                                                                 |

## EDUCATION

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| 2018 - 2022 | <b>Politecnico di Milano - Ph.D. in Machine Learning</b> <ul style="list-style-type: none"><li>• Focus on reinforcement learning, online planning, bandits, online convex optimization</li><li>• Applications: quantitative trading, optimal execution, market making, hedging, portfolio optimization</li></ul>                    |
| 2017 - 2018 | <b>SDA Bocconi, Milan - Executive Master in Finance</b> <ul style="list-style-type: none"><li>• Specialized in advanced financial topics with a focus on banking transformation and leadership</li></ul>                                                                                                                            |
| 2012 - 2016 | <b>Imperial College London - Master of Science in Mathematics, First</b> <ul style="list-style-type: none"><li>• Nominated Secretary of the Erasmus Society and External Relations Officer of the Italian Society</li><li>• Selected for Microfinance Tanzania, Mash Foundation; volunteered by advising local businesses</li></ul> |
| 2014 - 2015 | <b>EPFL, Lausanne, Switzerland - Exchange year, Swiss-European Mobility Program</b> <ul style="list-style-type: none"><li>• Engaged in Harvard World MUN (Model United Nations) debate, Seoul</li></ul>                                                                                                                             |

## PUBLICATIONS

- Optimal Execution with Reinforcement Learning. *arXiv*, 2024
- Exploiting Risk-Aversion and Size-dependent fees in FX Trading with Fitted Natural Actor-Critic. *arXiv*, 2024
- CVA Hedging with Reinforcement Learning. *ICAIF*, 2023
- Dealer markets: a reinforcement learning mean field game approach. *North American Journal of Economics and Finance*, 2023
- Dark-Pool Smart Order Routing: a Combinatorial Multi-armed Bandit Approach. *ICAIF*, 2022
- Addressing Non-Stationarity in FX Trading with Online Model Selection of Offline RL Experts. *ICAIF*, 2022
- MCTS for Trading and Hedging. *ICAIF*, 2021
- Learning FX Trading Strategies with FQI and Persistent Actions. *ICAIF*, 2021
- Conservative Online Convex Optimization. *ECML*, 2021
- Dealing with Transaction Costs in Portfolio Optimization: OGDM. *ICAIF*, 2020
- Option Hedging with Risk Averse Reinforcement Learning. *ICAIF*, 2020
- Risk-Averse Trust Region Optimization for Reward-Volatility Reduction. *IJCAI*, 2020

## LANGUAGES

Bilingual in ITALIAN and ENGLISH    Fluent in FRENCH

## PROGRAMMING SKILLS

Advanced PYTHON    Intermediate VBA, MATLAB, R, SQL

## INVOLVEMENT AND UPSKILLING

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|-------------|-----------------------------------------------------------------------------------------------|
| 2023-2024   | Started a fintech startup that simplifies finances for startups. Accelerated at Bocconi's B4i |
| SUMMER 2021 | DeepTech Entrepreneurship School   Milan, Italy                                               |
| SUMMER 2020 | Deep Learning Reinforcement Learning Summer School   Montreal, Canada                         |
| SUMMER 2015 | Alpbach Astrophysics Summer School   Alpbach, Austria                                         |